16. A complex in which the oxidation number of the metal is zero is

a) $K_4\left[Fe(CN)_6\right]$ b) $\left[Fe(CN)_3(NH_3)_3\right]$

c) $\left[Fe(CO)_5\right]$ d) both (b) and (c)

17. Formula of tris(ethane-1,2-diamine)iron(II)phosphate

a) $\left[Fe(CH_3\text{-}CH(NH_2)_2)_3\right](PO_4)_3$ b) $\left[Fe(H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2)_3\right](PO_4)$

c) $\left[Fe(H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2)_3\right](PO_4)_2$ d) $\left[Fe(H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2)_3\right]_3(PO_4)_2$

18. Which of the following is paramagnetic in nature?

a) $\left[Zn(NH_3)_4\right]^{2+}$ b) $\left[Co(NH_3)_6\right]^{3+}$

c) $\left[Ni(H_2O)_6\right]^{2+}$ d) $\left[Ni(CN)_4\right]^{2-}$

19. Fac-mer isomerism is shown by

a) $\left[Co(en)_3\right]^{3+}$ b) $\left[Co(NH_3)_4(Cl)_2\right]^{+}$

c) $\left[Co(NH_3)_3(Cl)_3\right]$ d) $\left[Co(NH_3)_5Cl\right]SO_4$

20. Choose the correct statement.

a) Square planar complexes are more stable than octahedral complexes

b) The spin only magnetic moment of $\left[Cu(Cl)_4\right]^{2-}$ is 1.732 BM and it has square planar structure.

c) Crystal field splitting energy (Δ_0) of $[FeF_6]^{4-}$ is higher than the (Δ_0) of $\left[Fe(CN)_6\right]^{4-}$

d) crystal field stabilization energy of $\left[V(H_2O)_6\right]^{2+}$ is higher than the crystal field stabilization of $\left[Ti(H_2O)_6\right]^{2+}$

Answer the following questions:

1. Write the IUPAC names for the following complexes.

i) $Na_2\left[Ni(EDTA)\right]$

ii) $\left[Ag(CN)_2\right]^-$

iii) $\left[Co(en)_3\right]_2(SO_4)_3$

iv) $\left[Co(ONO)(NH_3)_5\right]^{2+}$

v) $\left[Pt(NH_3)_2Cl(NO_2)\right]$

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